

Tutorial #6

Topics covered:

- Topologies on product spaces and continuity
- More about metric spaces

Problem 1

Consider the product spaces $\prod X_\alpha$ and $\prod Y_\alpha$ in the box topologies over the index set J . Prove that if $f : \prod X_\alpha \rightarrow \prod Y_\alpha$ defined by

$$f((x_\alpha)_{\alpha \in J}) = (f_\alpha(x_\alpha))_{\alpha \in J}$$

and each $f_\alpha : X_\alpha \rightarrow Y_\alpha$ is continuous, then f is continuous. Additionally:

- (i) Does this conclusion hold true when $\prod X_\alpha$ and $\prod Y_\alpha$ are endowed with the product topologies? Justify your answer.
- (ii) Does this conclusion hold true when $\prod X_\alpha$ is endowed with the box topology and $\prod Y_\alpha$ is endowed with the product topology? Justify your answer.
- (iii) Does this conclusion hold true when $\prod X_\alpha$ is endowed with the product topology and $\prod Y_\alpha$ is endowed with the box topology? Justify your answer.

Problem 2

Given sequences $(x_1, x_2, \dots)$ and $(b_1, b_2, \dots)$ of real numbers with $a_i > 0$ for all $i \in \mathbb{N}$, define $h : \mathbb{R}^\omega \rightarrow \mathbb{R}^\omega$ by the equation

$$h((x_1, x_2, \dots)) = (a_1x_1 + b_1, a_2x_2 + b_2, \dots).$$

Show that if $\mathbb{R}^\omega$ is given the product topology, h is a homeomorphism of $\mathbb{R}^\omega$ with itself. What happens if $\mathbb{R}^\omega$ is given the box topology?

Problem 3

Let X be metric space with metric d .

- (a) Show that $d : X \times X \rightarrow \mathbb{R}$ is continuous.
- (b) Let X' denote a space having the same underlying set as X . Show that if $d : X' \times X' \rightarrow \mathbb{R}$ is continuous, then the topology of X' is finer than the topology of X .

One can summarize the result of this exercise as follows: If X has a metric d , then the topology induced by d is the coarsest topology relative to which the function d is continuous.

Problem 4

Suppose that X is a topological space and Y and Y' are topological spaces on the same set, and that Y' is finer than Y . Suppose also that $f : X \rightarrow Y$ so that of course it is also a function from X to Y' . Prove the following statements:

- (1) If f is continuous with respect to Y' then it is also continuous with respect to Y .
- (2) If f is not continuous with respect to Y then it is also not continuous with respect to Y' .
- (3) If a sequence in Y' converges to a point y_0 , then it also converges to y_0 in Y .
- (4) If a sequence in Y does not converge to a point y_0 , then it also does not converge to y_0 in Y' .
- (5) If a sequence in Y does not converge at all, then it also does not converge at all in Y' .
- (6) If Y is a Hausdorff space, then so is Y' .

Problem 5

Show that if d is a metric for X , then

$$d'(x, y) = d(x, y)/(1 + d(x, y))$$

is a bounded metric that gives the same topology of X .

Hint: If $f(x) = x/(1+x)$ for $x > 0$, use the mean value theorem to show that $f(a+b) - f(b) \leq f(a)$.

Problem 6

Let X and Y be metric spaces with metrics d_X and d_Y , respectively. Let $f : X \rightarrow Y$ have the property that for every pair of points x_1, x_2 of X ,

$$d_Y(f(x_1), f(x_2)) = d_X(x_1, x_2).$$

Show that f is an imbedding. It is called an isometric imbedding of X in Y .